Traceable Hybrid Auditing of Clinical LLM Outputs: A Proof-of-Concept Combining Span Verification, Protocol Rules, and Graph Reasoning Kernels

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Abstract

Background: Evaluating Large Language Models (LLMs) in open-ended clinical workflows such as acute chest pain triage requires auditing procedural adherence (omissions, temporal ordering, and contraindications) rather than static multiple-choice accuracy. Unconstrained *LLM-as-a-Judge* paradigms risk fabricated event extraction and often penalize appropriate diagnostic uncertainty.

Objective: We present a proof-of-concept **Traceable Neuro-Symbolic Auditing Framework** and evaluate its mechanism calibration on a pilot benchmark ($N = 10$ traces, plus $N = 5$ linguistic edge cases).

Methods: The architecture decouples auditing into three layers: (1) a `TraceExtractor` enforcing deterministic verbatim span verification and epistemic assertion calibration (`PROPOSED`, `REPORTED_PERFORMED`, `NEGATED`, `SUSPECTED`, `DEFINITIVE`), (2) a `ClinicalRuleEvaluator` checking temporal ordering and contraindications, and (3) a graph-based `Reasoning Kernel` detecting unsupported definitive claims (`Pattern RP-001`).

Pilot Results: Across our 10-case pilot calibration suite (5 synthetic failure/control traces and 5 real `GPT-4o` responses), introducing assertion-status calibration eliminated 3 false-positive flags on negated actions and suspected differentials, shifting pilot specificity from 4/7 (57.1%, 95% CI: 18.4%–90.1%) to 7/7 (100.0%, 95% CI: 59.0%–100.0%) while preserving 3/3 sensitivity on synthetic protocol violations.

1 Introduction

While Large Language Models (LLMs) perform strongly on static medical licensing examinations [4], open-ended emergency triage of **Acute Chest Pain** [1] requires strict procedural adherence. An LLM may list a correct differential diagnosis while proposing lethal management steps, such as discharging a patient after symptomatic relief from a gastrointestinal cocktail prior to reviewing serial troponins (premature closure), or administering empiric heparin in suspected aortic dissection.

Current *LLM-as-a-Judge* evaluation methods [5] face two structural limitations in procedural clinical auditing:

1. **Unverified Event Extraction:** A neural judge may hallucinate that a candidate response ordered an investigation when no such statement exists in the text.
2. **Epistemic Conflation:** Inspired by classical clinical negation and context detection algorithms such as `NegEx` [2] and `ConText` [3], an effective auditor must distinguish negated warnings (*"Do not give heparin"*) and suspected differentials from definitive clinical commitments.

2 System Architecture

Our framework separates linguistic extraction from deterministic symbolic verification across three tiers:

2.1 Tier 1: `TraceExtractor` (Span Firewall & Epistemic Tagging)

Given raw text T, candidate events $E = \{e_1, \ldots, e_n\}$ must satisfy verbatim substring verification under whitespace/case normalization $\mathcal{N}(s_i) \subseteq \mathcal{N}(T)$, rejecting ungrounded extractions (`SPAN_NOT_IN_SOURCE`). Lexical synonyms are mapped to canonical protocol codes, and events are filtered by assertion status $\alpha(e_i)$ so that `NEGATED` warnings are never penalized as proposed actions, and only `DEFINITIVE` inferences require structural evidentiary proof.

2.2 Tier 2 & Tier 3: `ClinicalRuleEvaluator` & `Reasoning Kernel`

The rule engine evaluates mandatory omissions (`CRITICAL_OMISSION`), prerequisite temporal ordering ($a_{\text{pre}} \prec a_{\text{target}}$) for premature closure, and contraindications (`must_avoid`). Concurrently, the `Reasoning Kernel` constructs a Directed Acyclic Graph $G = (V, E_{\text{graph}})$ and triggers `RP-001` whenever a `DEFINITIVE` inference node has support in-degree $\deg^-_{\text{SUPPORTS}}(v) = 0$.

3 Pilot Calibration & Mechanism Verification

Table 1 reports the pilot mechanism ablation across $N = 10$ calibration traces (3 synthetic protocol violations, 2 synthetic safe traces, and 5 real `GPT-4o` responses) alongside an unseen 5-case linguistic stress suite (`run_edge_cases.py`).

Table 1: Pilot Mechanism Ablation: Assertion Calibration on $N = 10$ Traces

Auditor Configuration	TP	FP	TN	FN	Recall (3/3)	Specificity (TN/7)	Pilot F_1
1. Uncalibrated Baseline	3	3	4	0	100.0% (29.2–100%)	57.1% (18.4–90.1%)	0.667
2. Calibrated Hybrid Pipeline	**3**	**0**	**7**	**0**	**100.0% (29.2–100%)**	**100.0% (59.0–100%)**	**1.000**

4 Methodological Limitations & Future Scaling

This work represents a software architecture and pilot calibration study rather than a large-scale empirical validation. Specifically: (1) the sample size ($N = 10$ calibration traces, $N = 5$ edge cases) yields wide Clopper-Pearson 95% confidence intervals; (2) rules and synthetic test cases were authored by a single investigator without independent multi-clinician inter-annotator agreement (κ); and (3) while substring verification confirms quote existence, upstream assertion classification remains dependent on parser accuracy. Future work will expand evaluation to > 50 multi-disease scenarios with blinded dual-clinician annotation and direct comparison against standalone LLM judges.

References

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